A variant of Kemperman's theorem

Terence Tao · 26 December 2011

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In 1964, Kemperman established the following result:

> **Theorem 1** Let G be a compact connected group, with a Haar probability measure μ. Let A, B be compact subsets of G. Then
>
> $$\mu(AB) \geq \min(\mu(A) + \mu(B), 1).$$

> **Remark 1** The estimate is sharp, as can be seen by considering the case when G is a unit circle, and A, B are arcs; similarly if G is any compact connected group that projects onto the circle. The connectedness hypothesis is essential, as can be seen by considering what happens if A and B are a non-trivial open subgroup of G. For locally compact connected groups which are unimodular but not compact, there is an analogous statement, but with μ now a Haar measure instead of a Haar probability measure, and the right-hand side $\min(\mu(A) + \mu(B), 1)$ replaced simply by $\mu(A) + \mu(B)$. The case when G is a torus is due to Macbeath, and the case when G is a circle is due to Raikov. The theorem is closely related to the Cauchy-Davenport inequality; indeed, it is not difficult to use that inequality to establish the circle case, and the circle case can be used to deduce the torus case by considering increasingly dense circle subgroups of the torus (alternatively, one can also use Kneser's theorem).
>
> By inner regularity, the hypothesis that A, B are compact can be replaced with Borel measurability, so long as one then adds the additional hypothesis that $A + B$ is also Borel measurable.

A short proof of Kemperman's theorem was given by Ruzsa. In this post I wanted to record how this argument can be used to establish the following more "robust" version of Kemperman's theorem, which not only lower bounds AB, but gives many elements of AB some multiplicity:

> **Theorem 2** Let G be a compact connected group, with a Haar probability measure μ. Let A, B be compact subsets of G. Then for any $0 \leq t \leq \min(\mu(A), \mu(B))$, one has
>
> $$\int_G \min(1_A * 1_B, t)\, d\mu \geq t \min(\mu(A) + \mu(B) - t, 1). \tag{1}$$

Indeed, Theorem 1 can be deduced from Theorem 2 by dividing (1) by t and then taking limits as $t \to 0$. The bound in (1) is sharp, as can again be seen by considering the case when A, B are arcs in a circle. The analogous claim for cyclic groups for prime order was established by Pollard, and for general abelian groups by Green and Ruzsa.

Let us now prove Theorem 2. It uses a submodularity argument related to one discussed in this previous post. We fix B and t with $0 \leq t \leq \mu(B)$, and define the quantity

$$c(A) := \int_G \min(1_A * 1_B, t)\, d\mu - t(\mu(A) + \mu(B) - t).$$

for any compact set A. Our task is to establish that $c(A) \geq 0$ whenever $t \leq \mu(A) \leq 1 - \mu(B) + t$.

We first verify the extreme cases. If $\mu(A) = t$, then $1_A * 1_B \leq t$, and so $c(A) = 0$ in this case (since $\int_G 1_A * 1_B = \mu(A)\mu(B) = t\mu(B)$). At the other extreme, if $\mu(A) = 1 - \mu(B) + t$, then from the inclusion-exclusion principle we see that $1_A * 1_B \geq t$, and so again $c(A) = 0$ in this case.

To handle the intermediate regime when $\mu(A)$ lies between t and $1 - \mu(B) + t$, we rely on the *submodularity inequality*

$$c(A_1) + c(A_2) \geq c(A_1 \cap A_2) + c(A_1 \cup A_2) \tag{2}$$

for arbitrary compact A_1, A_2. This inequality comes from the obvious pointwise identity

$$1_{A_1} + 1_{A_2} = 1_{A_1 \cap A_2} + 1_{A_1 \cup A_2}$$

whence

$$1_{A_1} * 1_B + 1_{A_2} * 1_B = 1_{A_1 \cap A_2} * 1_B + 1_{A_1 \cup A_2} * 1_B$$

and thus (noting that the quantities on the left are closer to each other than the quantities on the right)

$$\min(1_{A_1} * 1_B, t) + \min(1_{A_2} * 1_B, t)$$

$$\geq \min(1_{A_1 \cap A_2} * 1_B, t) + \min(1_{A_1 \cup A_2} * 1_B, t)$$

at which point (2) follows by integrating over G and then using the inclusion-exclusion principle.

Now introduce the function

$$f(a) := \inf\{c(A) : \mu(A) = a\}$$

for $t \leq a \leq 1 - \mu(B) + t$. From the preceding discussion $f(a)$ vanishes at the endpoints $a = t, 1 - \mu(B) + t$; our task is to show that $f(a)$ is non-negative in the interior region $t < a < 1 - \mu(B) + t$. Suppose for contradiction that this was not the case. It is easy to see that f is continuous (indeed, it is even Lipschitz continuous), so there must be $t < a < 1 - \mu(B) + t$ at which f is a local minimum and not locally constant. In particular, $0 < a < 1$. But for any A with $\mu(A) = a$, we have the translation-invariance

$$c(gA) = c(A) \tag{3}$$

for any $g \in G$, and hence by (2)

$$c(A) \geq \frac{1}{2} c(A \cap gA) + \frac{1}{2} c(A \cup gA).$$

Note that $\mu(A \cap gA)$ depends continuously on g, equals a when g is the identity, and has an average value of a^2. As G is connected, we thus see from the intermediate value theorem that for any $0 < \epsilon < a - a^2$, we can find g such that $\mu(A \cap gA) = a - \epsilon$, and thus by inclusion-exclusion $\mu(A \cup gA) = a + \epsilon$. By definition of f, we thus have

$$c(A) \geq \frac{1}{2} f(a - \epsilon) + \frac{1}{2} f(a + \epsilon).$$

Taking infima in A (and noting that the hypotheses on ϵ are independent of A) we conclude that

$$f(a) \geq \frac{1}{2} f(a - \epsilon) + \frac{1}{2} f(a + \epsilon)$$

for all $0 < \epsilon < a - a^2$. As f is a local minimum and ϵ is arbitrarily small, this implies that f is locally constant, a contradiction. This establishes Theorem 2.

We observe the following corollary:

> **Corollary 3** Let G be a compact connected group, with a Haar probability measure μ. Let A, B, C be compact subsets of G, and let $\delta := \min(\mu(A), \mu(B), \mu(C))$. Then one has the pointwise estimate
>
> $$1_A * 1_B * 1_C \geq \frac{1}{4}(\mu(A) + \mu(B) + \mu(C) - 1)_+^2$$
>
> if $\mu(A) + \mu(B) + \mu(C) - 1 \leq 2\delta$, and
>
> $$1_A * 1_B * 1_C \geq \delta(\mu(A) + \mu(B) + \mu(C) - 1 - \delta)$$
>
> if $\mu(A) + \mu(B) + \mu(C) - 1 \geq 2\delta$.

Once again, the bounds are completely sharp, as can be seen by computing $1_A * 1_B * 1_C$ when A, B, C are arcs of a circle. For quasirandom G, one can do much better than these bounds, as discussed in this recent blog post; thus, the abelian case is morally the worst case here, although it seems difficult to convert this intuition into a rigorous reduction.

Proof: By cyclic permutation we may take $\delta = \mu(C)$. For any

$$(\mu(A) + \mu(B) - 1)_+ \leq t \leq \min(\mu(A), \mu(B)),$$

we can bound

$$\begin{aligned} 1_A * 1_B * 1_C &\geq \min(1_A * 1_B, t) * 1_C \\ &\geq \int_G \min(1_A * 1_B, t)\, d\mu - t(1 - \mu(C)) \\ &\geq t(\mu(A) + \mu(B) - t) - t(1 - \mu(C)) \\ &= t \min(\mu(A) + \mu(B) + \mu(C) - 1 - t) \end{aligned}$$

where we used Theorem 2 to obtain the third line. Optimising in t, we obtain the claim. □

Comments (7)

 **Anonymous** · 26 December 2011

Dear Terry ,
From the estimate you explained for \mu(AB) (A, B are two compact subset of the group G) I am wondering that when the distribution $1_A \star 1_B$ is absolutely continuous with respect to the Haar measure of the group G.
Also what is the least integer number n=n(A, B) for which the power measure $(1_A \star 1_B)^n$ is absolutely continuous ? I think This integer number has interesting property which depend on the sets A and B.

 **pavel zorin** · 27 December 2011

Dear Terence,

the inequality in the proof of (2) is probably reversed.

It might also be worth mentioning that f(a) is in fact Lipschitz to avoid people like me following a wrong lead. :)

best regards,
pavel

[Corrected, thanks – T.]

 **Sean Eberhard** · 14 November 2012

Dear Terry,

To what extent can one generalise this argument to not-necessarily-connected groups? Can one get a more robust version of Kneser's theorem (in a locally compact group), or at least an alternative proof of Kneser's theorem?

Best wishes,
Sean

 **Terence Tao** · 14 November 2012

That's a good question. I know some people have already tried to use submodularity methods to prove Kneser's theorem, but it seems to be a surprisingly difficult task; it seems to be quite difficult to hit upon a way of iterating submodularity so that one arrives at a trivially true inequality in finite time (as is the case for connected groups as outlined above).

Certainly I would like to see a new proof of that theorem, as the current proofs are quite involved...

 **Sean Eberhard** · 31 July 2013

Tiny correction: the parenthetical remark "since $\int_G 1_A * 1_B = \mu(A)\mu(B) = t\mu(B)$" just before equation (2) is attached to the wrong sentence, which confused me briefly.

[Corrected, thanks – T.]

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